

Rémi PYT
Antoine RICHARD
Matthieu Sirier
Jérémy SAGNARD

Project Description

ArtConnect Application

Step 2: Conceptual and Logical Modeling of the ArtConnect Database

Objective: Design the conceptual and logical models of the database from the identified features, applying normalization principles (up to 3NF).

Tasks:

We directly put the identifiers inside our entities as well as for our functional dependencies.

1. Identify entities and relationships

- Based on the interface and business needs, identify the different entities and their attributes.

The different entities are:

Underscored attributes correspond to arrays, they will be later implemented as a relationship to follow atomicity in the attributes.

Galleries	Exhibition	Artwork	Artist
Gallery_Id Gallery_name Gallery_address Gallery_ownerName Gallery_openingHour Gallery_contactPhone Gallery_website Gallery_rating exhibitions	Exhibition_Id Exhibition_Title Exhibition_startDate Exhibition_endDate Exhibition_description gallery Exhibition_curatorName Exhibition_theme Exhibition_artworks	Artwork_Id Artwork_Exhibition_title Artwork_creationTear Artwork_type Artwork_medium Artwork_dimensions Artwork_description Artwork_price Artwork_status artist Artwork_tags	Artist_Id Artist_name Artist_bio Artist_birthyear Artist_disciplines Artist_contactEmail Artist_phone Artist_city Artist_website Artist_socialMedia Artist_isActive artworks
Workshop	Review	Booking	CommunityMember
Workshop_Id Workshop_title Workshop_date Workshop_surationMinutes Workshop_maxParticipants Workshop_price instructor Workshop_location Workshop_description Workshop_level_Work	Review_Reviewer artwork Review_rating Review_comment Review_reviewDate Review_type	Workshop Member Booking_bookingDate Booking_paymentStatus	CM_Id CM_Name CM_email CM_birthYear CM_phone CM_city CM_favoriteDisciplines CM_membershipType CM_bookings CM_reviews

- Define the functional dependencies between the different attributes.

Galleries:

gallery_id -> Gallery_name, Gallery_Address, Gallery_ownerName, Gallery_openingHour,
Gallery_contactPhone, Gallery_website, Gallery_rating, exhibition

Exhibitions:

exhibition_id, gallery_id -> Exhibition_title, Exhibition_startDate, Exhibition_endDate,
Exhibition_description, Exhibition_curatorName, Exhibition_theme, artworks

Artwork:

Artwork_id -> Artwork_title, Artwork_creationTear, Artwork_type, Artwork_medium,
Artwork_dimensions, Artwork_description, Artwork_price, Artwork_status, artist, Artwork_tags

Artist:

Artist_id -> Artist_name, Artist_bio, Artist_birthyear, Artist_disciplines, Artist_contactEmail,
Artist_phone, Artist_city, Artist_website, Artist_socialMedia, Artist_isActive, artworks

Workshop:

Workshop_id -> Workshop_title, Workshop_date, Workshop_durationMinutes,
Workshop_maxParticipants, Workshop_price, instructor, Workshop_location,
Workshop_description, Workshop_level

Review:

reviewer, artwork -> Review_rating, Review_comment, Review_reviewDate, Review_type

Booking:

Workshop_id, member_id -> Booking_bookingDate, Booking_paymentStatus

CommunityMember:

Member_id -> CM_Name, CM_email, CM_birthyear, CM_phone, CM_city, CM_favoriteDisciplines,
CM_membershipType, CM_bookings, CM_reviews

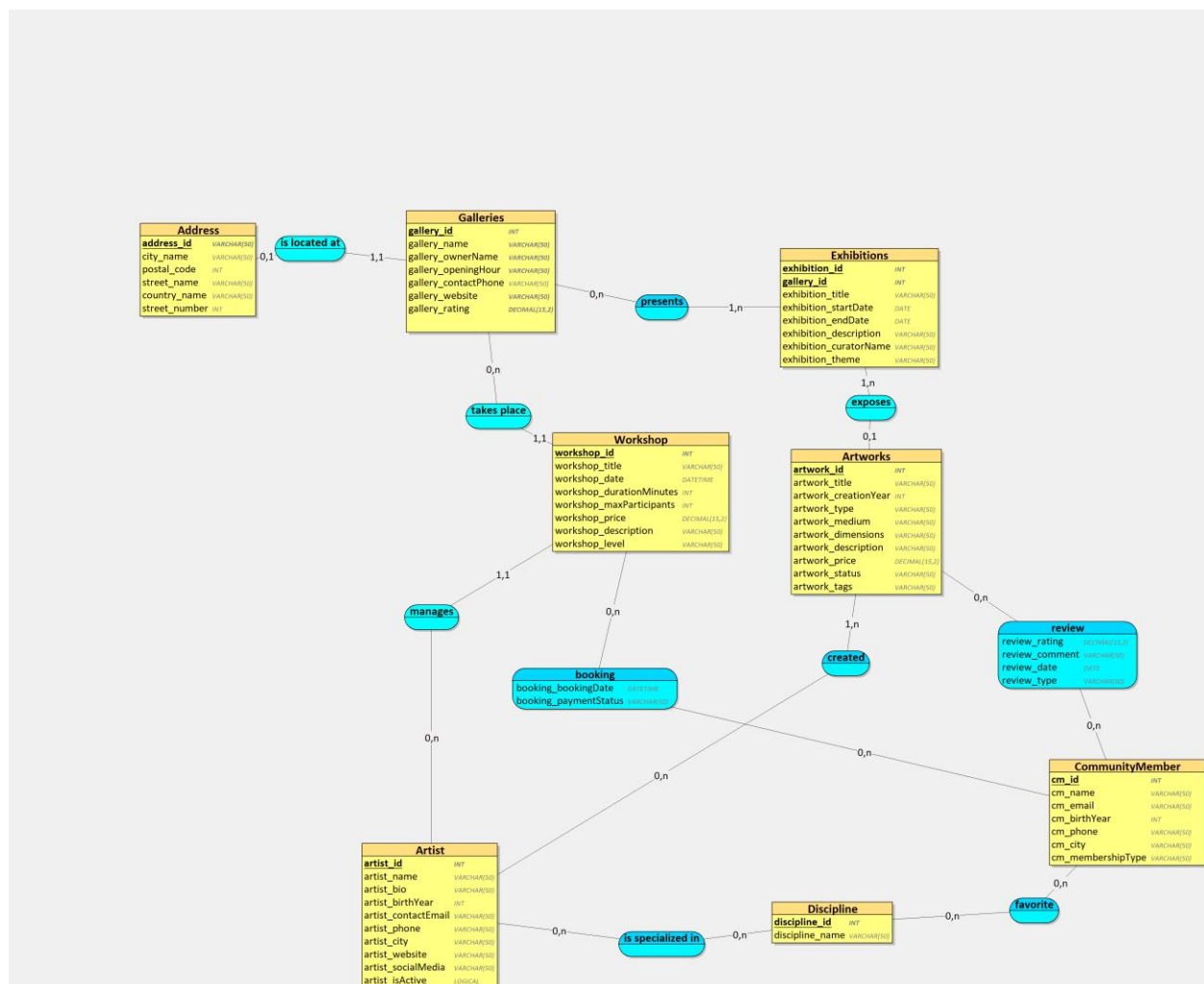
Database 2 : Security & Automation

- Think about the identifiers to define for each entity. You can choose one (or more) existing attribute(s) from the associated class, or define a new one.

Entity Name	ID
Galleries	Gallery_id
Exhibitions	Exhibition_id, gallery_id
Artwork	Artwork_id
Artist	Artist_id
Workshop	Workshop_id
Review	Review_id
Booking	Workshop_id, member_id
CommunityMember	Member_id

2. Create a Conceptual Data Model (CDM)

- Use an Entity-Relationship diagram (ERD) to represent entities, attributes, and relationships.



3. Create a Logical Data Model (LDM)

- Transform the CDM into an LDM by adding primary keys, foreign keys, and data types.
- Normalize the database up to the 3rd Normal Form (3NF).

Exhibitions = (exhibition_id INT, gallery_id INT, exhibition_title VARCHAR(50), exhibition_startDate DATE, exhibition_endDate DATE, exhibition_description VARCHAR(50), exhibition_curatorName VARCHAR(50), exhibition_theme VARCHAR(50));

Artworks = (artwork_id INT, artwork_title VARCHAR(50), artwork_creationYear INT, artwork_type VARCHAR(50), artwork_medium VARCHAR(50), artwork_dimensions VARCHAR(50), artwork_description VARCHAR(50), artwork_price DECIMAL(15,2), artwork_status VARCHAR(50), artwork_tags VARCHAR(50), #(exhibition_id, gallery_id)*);

Artist = (artist_id INT, artist_name VARCHAR(50), artist_bio VARCHAR(50), artist_birthYear INT, artist_contactEmail VARCHAR(50), artist_phone VARCHAR(50), artist_city VARCHAR(50), artist_website VARCHAR(50), artist_socialMedia VARCHAR(50), artist_isActive LOGICAL);

CommunityMember = (cm_id INT, cm_name VARCHAR(50), cm_email VARCHAR(50), cm_birthYear INT, cm_phone VARCHAR(50), cm_city VARCHAR(50), cm_membershipType VARCHAR(50));

Discipline = (discipline_id INT, discipline_name VARCHAR(50));

Address = (address_id VARCHAR(50), city_name VARCHAR(50), postal_code INT, street_name VARCHAR(50), country_name VARCHAR(50), street_number INT);

Galleries = (gallery_id INT, gallery_name VARCHAR(50), gallery_ownerName VARCHAR(50), gallery_openingHour VARCHAR(50), gallery_contactPhone VARCHAR(50), gallery_website VARCHAR(50), gallery_rating DECIMAL(15,2), #address_id);

Workshop = (workshop_id INT, workshop_title VARCHAR(50), workshop_date DATETIME, workshop_durationMinutes INT, workshop_maxParticipants INT, workshop_price DECIMAL(15,2), workshop_description VARCHAR(50), workshop_level VARCHAR(50), #artist_id, #gallery_id);

created = (#artwork_id, #artist_id);

presents = (#gallery_id, #(exhibition_id, gallery_id 1));

booking = (#workshop_id, #cm_id, booking_bookingDate DATETIME, booking_paymentStatus VARCHAR(50));

is_specialized_in = (#artist_id, #discipline_id);

favorite = (#cm_id, #discipline_id);

review = (#artwork_id, #cm_id, review_rating DECIMAL(15,2), review_comment VARCHAR(50), review_date DATE, review_type VARCHAR(50));

4. Create the database schema

- Write SQL scripts to create tables, define constraints (e.g., primary keys, foreign keys, NOT NULL), and establish relationships.

```
CREATE TABLE Exhibitions(  
    exhibition_id INT,  
    gallery_id INT,  
    exhibition_title VARCHAR(50),  
    exhibition_startDate DATE,  
    exhibition_endDate DATE,  
    exhibition_description VARCHAR(50),  
    exhibition_curatorName VARCHAR(50),  
    exhibition_theme VARCHAR(50),  
    PRIMARY KEY(exhibition_id, gallery_id)  
);  
  
CREATE TABLE Artworks(  
    artwork_id INT,  
    artwork_title VARCHAR(50),  
    artwork_creationYear INT,  
    artwork_type VARCHAR(50),  
    artwork_medium VARCHAR(50),  
    artwork_dimensions VARCHAR(50),  
    artwork_description VARCHAR(50),  
    artwork_price DECIMAL(15,2),  
    artwork_status VARCHAR(50),  
    artwork_tags VARCHAR(50),  
    exhibition_id INT,  
    gallery_id INT,  
    PRIMARY KEY(artwork_id),  
    FOREIGN KEY(exhibition_id, gallery_id) REFERENCES  
Exhibitions(exhibition_id, gallery_id)  
);  
  
CREATE TABLE Artist(  
    artist_id INT,  
    artist_name VARCHAR(50),  
    artist_bio VARCHAR(50),  
    artist_birthYear INT,  
    artist_contactEmail VARCHAR(50),  
    artist_phone VARCHAR(50),  
    artist_city VARCHAR(50),  
    artist_website VARCHAR(50),  
    artist_socialMedia VARCHAR(50),  
    artist_isActive LOGICAL,  
    PRIMARY KEY(artist_id)  
);  
  
CREATE TABLE CommunityMember(  
    cm_id INT,  
    cm_name VARCHAR(50),  
    cm_email VARCHAR(50),  
    cm_birthYear INT,
```

```

cm_phone VARCHAR(50),
cm_city VARCHAR(50),
cm_membershipType VARCHAR(50),
PRIMARY KEY(cm_id)
);

CREATE TABLE Discipline(
    discipline_id INT,
    discipline_name VARCHAR(50),
    PRIMARY KEY(discipline_id)
);

CREATE TABLE Address(
    address_id VARCHAR(50),
    city_name VARCHAR(50),
    postal_code INT,
    street_name VARCHAR(50),
    country_name VARCHAR(50),
    street_number INT,
    PRIMARY KEY(address_id)
);

CREATE TABLE Galleries(
    gallery_id INT,
    gallery_name VARCHAR(50) NOT NULL,
    gallery_ownerName VARCHAR(50) NOT NULL,
    gallery_openingHour VARCHAR(50) NOT NULL,
    gallery_contactPhone VARCHAR(50),
    gallery_website VARCHAR(50) NOT NULL,
    gallery_rating DECIMAL(15,2) NOT NULL,
    address_id VARCHAR(50) NOT NULL,
    PRIMARY KEY(gallery_id),
    UNIQUE(address_id),
    FOREIGN KEY(address_id) REFERENCES Address(address_id)
);

CREATE TABLE Workshop(
    workshop_id INT,
    workshop_title VARCHAR(50),
    workshop_date DATETIME,
    workshop_durationMinutes INT,
    workshop_maxParticipants INT,
    workshop_price DECIMAL(15,2),
    workshop_description VARCHAR(50),
    workshop_level VARCHAR(50),
    artist_id INT NOT NULL,
    gallery_id INT NOT NULL,
    PRIMARY KEY(workshop_id),
    FOREIGN KEY(artist_id) REFERENCES Artist(artist_id),
    FOREIGN KEY(gallery_id) REFERENCES Galleries(gallery_id)
);

CREATE TABLE created(
    artwork_id INT,
    artist_id INT,
    PRIMARY KEY(artwork_id, artist_id),

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```

FOREIGN KEY(artwork_id) REFERENCES Artworks(artwork_id),
FOREIGN KEY(artist_id) REFERENCES Artist(artist_id)
);

CREATE TABLE presents(
    gallery_id INT,
    exhibition_id INT,
    gallery_id_1 INT,
    PRIMARY KEY(gallery_id, exhibition_id, gallery_id_1),
    FOREIGN KEY(gallery_id) REFERENCES Galleries(gallery_id),
    FOREIGN KEY(exhibition_id, gallery_id_1) REFERENCES
Exhibitions(exhibition_id, gallery_id)
);

CREATE TABLE booking(
    workshop_id INT,
    cm_id INT,
    booking_bookingDate DATETIME,
    booking_paymentStatus VARCHAR(50),
    PRIMARY KEY(workshop_id, cm_id),
    FOREIGN KEY(workshop_id) REFERENCES Workshop(workshop_id),
    FOREIGN KEY(cm_id) REFERENCES CommunityMember(cm_id)
);

CREATE TABLE is_specialized_in(
    artist_id INT,
    discipline_id INT,
    PRIMARY KEY(artist_id, discipline_id),
    FOREIGN KEY(artist_id) REFERENCES Artist(artist_id),
    FOREIGN KEY(discipline_id) REFERENCES Discipline(discipline_id)
);

CREATE TABLE favorite(
    cm_id INT,
    discipline_id INT,
    PRIMARY KEY(cm_id, discipline_id),
    FOREIGN KEY(cm_id) REFERENCES CommunityMember(cm_id),
    FOREIGN KEY(discipline_id) REFERENCES Discipline(discipline_id)
);

CREATE TABLE review(
    artwork_id INT,
    cm_id INT,
    review_rating DECIMAL(15,2),
    review_comment VARCHAR(50),
    review_date DATE,
    review_type VARCHAR(50),
    PRIMARY KEY(artwork_id, cm_id),
    FOREIGN KEY(artwork_id) REFERENCES Artworks(artwork_id),
    FOREIGN KEY(cm_id) REFERENCES CommunityMember(cm_id)
);

```